MIL-M-60157C(MU)
20 October 1972
SUPERSEDING
MIL-M-60157B(MU)
10 July 1969

MILITARY SPECIFICATION

MOUNT, TELESCOPE: M149

1. SCOPE

1.1 This specification covers one type of telescope mount which supports Telescopes M119 and M127 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

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STANDARDS (Cont'd)

Military

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army Munitions Command

F8565556	Pressure Tester - Two Station
F10527950	Mount, Telescope: M149
F10541523	Gage, Functional, Maximum
F10541525	Gage, Functional, Maximum

PACKAGING DATA SHEET

10527950	Packaging of Mount, Telescope: M149

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The mount shall be manufactured in accordance with Drawing F10527950 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926.

MIL-F-13926.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship
(f) Soldering electrical connections

3.3 Environmental service conditions.

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3.3.1 Storage temperatures.- The mount shall show no evidence of physical failure, such as, cement separation, glass breakage, or casting failure after having been exposed and thermally stabilized at temperatures of -80° and +160° Fahrenheit (F).

3.3.2 Extreme operating temperatures.- The mount shall be operable (3.19 and 3.19.1) at temperatures of -65° to +150°F.

3.3.3 Standard ambient temperatures.- Unless otherwise specified, the mount shall meet the requirements of 3.4 through 3.21, at standard ambient temperatures between 60° and 90°F.

3.4 Vibration.- The mount shall show no evidence of physical failure due to being vibrated successively along three (3) mutually perpendicular axes. Vibration shall be a simple harmonic motion, the frequency varying either lineraly or logarithmically from 5 to 55 to 5 CPS in five minute cycles for a total of 15 minutes along each of the axes. The amplitude of vibration shall be maintained constant at 1/64 inch (1/32 inch total excursion). Following the vibration test the mount shall meet all subsequent requirements.

3.5 Cleanliness.

3.5.1 Dirt.- Inspection for dirt and foreign particles (classed as dirt) shall be made with the unaided eye from both ends of the mount by transmitted and reflected light. The maximum size of any visible particle on the glass surfaces shall be 1/8 inch. The total number of particles shall not exceed five (5). The total number of particles on any one (1) glass component shall not exceed two (2). Dirt particles smaller than 1/32 inch shall be ignored regardless of distribution.

3.5.2 Other defects.- There shall be no evidence of moisture, grease, fingerprints or fractures when viewing through either end of the mount by transmitted and reflected light.

3.6 Sealing.- The mount shall show no evidence of leakage in excess of 0.2 pound per square inch gage when subjected to an internal pressure of 5 ± 0.10 pound per square inch (psig) for a minimum of 2 hours.

3.7 Telescope and tracker mounting features.

3.7.1 Orientation.- The mount shall be positioned with the telescope mounting surface of the mount in a vertical plane and the locating keyway for the telescope at the six o'clock position.

3.7.2 Insertion and removal.- The mount shall permit insertion and removal of the type telescope and tracker which the mount is intended to support. The mount shall meet this requirement with the tracker inserted and removed while the telescope is positioned in the mount and with the telescope inserted and removed while the tracker is positioned and bolted securely in the mount.

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3.7.3 Rigidity.- The locking mechanism shall securely lock the type telescope which the mount is intended to support.

3.8 Collimation.

3.8.1 Telescope-to-tracker.- With the incident line of sight perpendicular to the telescope mounting surfaces and entering the entrance window, the two emerging lines of sight shall remain perpendicular to this same mounting surface within 2 mils and parallel to each other within 2 mils.

3.8.2 Checksight-to-tracker.- With the ALIGN lever in the full clockwise position (error wedge removed from line of sight counterclockwise) the image of the illuminated checksight beam emerging from the beam-splitter plate shall be centered on the tracker cavity axis within 0.25 mil.

3.9 Resolution.- The resolution on the telescope cavity axis shall be 6.0 seconds of arc or less when using a telescope with a two inch entrance pupil at one established focus setting.

3.10 Light transmission.- White light transmission to the telescope cavity axis shall be at least 75 percent.

3.11 Spherical power.- The spherical power shall not exceed 5.0 millidiopters using a telescope of two inch entrance pupil on the telescope axis.

3.12 Astigmatism.- Astigmatism shall not exceed 3.0 millidiopters using a telescope of two inch entrance pupil on the telescope axis.

3.13 Wedge deviation.- From the position established in 3.8.2 the conditions of 3.13.1 and 3.13.2 shall be met.

3.13.1 Line of sight deviation.- The line of sight of the checksight beam shall deviate by 0.39 ± 0.03 mil in azimuth and 0.39 ± 0.03 mil in elevation and shall be in the upper right quadrant (above the horizontal centerline and to the right of the vertical centerline) when viewed along the tracker axis, and the ERROR lever is in the full clockwise position.

3.13.2 Repeatability.- At no time during repeated operation of the ERROR lever shall the line of sight deviate less than 0.36 mil or more than 0.42 mil when the ERROR lever is in the full clockwise position.

3.14 Signal frequency.- Signal frequency derived from light pulsations of 2400 ± 24 pulses per second shall be indicated on the

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tracker axis (see 3.8.2) through the beam-splitter with the following prevalent conditions:

(a) Chopper motor energized with 115 ± 6 vac at a frequency of 400 ± 0.4 cps.
(b) ALIGN lever in maximum clockwise position.
(c) ERROR lever in maximum counterclockwise position.

3.15 Diaphragm adjustment.- From the position established in 3.8.2 the AZ and EL boresight knob shall meet the rotation conditions of 3.15.1 and 3.15.2.

3.15.1 Elevation.- The checksight beam shall be adjustable over a range of 3.75 ± 0.25 mils in elevation and 3.75 ± 0.25 mils in depression in a vertical plane by rotation of the elevation boresight knob.

3.15.2 Azimuth.- The checksight beam shall be adjustable over a range of 3.75 ± 0.25 mils to the left and 3.75 ± 0.25 mils to the right in a horizontal plane by rotation of the azimuth boresight knob.

3.16 Checksight lens.- The checksight lens shall be adjusted such that the image of the diaphragm circle of the checksight, when viewed at the tracker aperture opening shall appear to be at a range of 2000 meters ± 100 meters.

3.17 Irradiant intensity (infrared power).- From the position established in 3.8.2, the effective power of the checksight beam at the fundamental frequency of 3.14, output shall be $7.5 \times 10^{-9} \pm 1.5 \times 10^{-9}$ rms watts.

3.18 Projected checksight image beam.- From the position established in 3.8.2 the image of the checksight beam, when viewed along the tracker axis shall be circular and shall have a diameter of angular subtense of 1.0 + .00 minus (-) .05 mil.

3.19 Operability.- The telescope locking lever, the checksight levers (ERROR AND ALIGN), and the diaphragm adjusting screws, AZ and EL, shall be mechanically operable over their full excursion of travel. The torque required to latch the ALIGN lever (full clockwise position) shall be 11 ± 3 inch pounds.

3.19.1 Solenoid.- The porro prism shall be completely removed from the optical path of the tracker and the ALIGN lever shall move to maximum counterclockwise position when the solenoid is energized. With 18V dc the maximum current draw shall not exceed 1.33 amperes, and with 28V dc the maximum current draw shall not exceed 2.0 amperes.

3.20 Electrical.

3.20.1 Checksight illumination.- With the checksight lamp energized, the diaphragm shall appear clearly defined when viewed

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through the exit windows of the mount. The current draw of the lamp shall be 0.080 ± 0.002 amperes.

3.20.2 Circuitry.- The circuitry of the mount shall be in accordance with the electrical schematic drawings.

3.20.3 Dielectric strength.- There shall be no puncture of insulation, no sparkover, no decrease in dielectric strenth when the insulated areas of the circuitry are subjected to an alternating current (ac) root mean square (rms) potential of 500 volts plus twice the rated voltage at a frequency of 60 cps for a period of 15 seconds ± 5 seconds.

3.20.4 Insulation resistance.- Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 100 megohms with 500V dc applied.

3.21 Interior atmosphere.- The interior atmosphere of the optical section of the mount shall contain nitrogen having a dew point at least as low as -25°F and the internal pressure of the mount shall be 3 psi ± 0.5 psi.

3.22 Workmanship.- The workmanship requirements of MIL-F-13926 shall be adhered to. Exterior protective surface finishes shall be free of voids which may affect the protective qualities of the finish or appearance. Optical surfaces shall be free of scratches and blemishes which may affect their optical qualities or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 mounts plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from

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the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAPs, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
101. Vibration	3.4	4.6.3
102. Cleanliness	3.5	4.6.4
103. Sealing	3.6	4.6.5
104. Interior atmosphere	3.21	4.6.6
105. Insertion and removal	3.7.2	4.6.7

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MAJOR		REQUIREMENT	TEST PROCEDURE
106.	Rigidity	3.7.3	4.6.8
107.	Resolution	3.9	4.6.9
108.	Light transmission	3.10	4.6.10
109.	Astigmatism	3.12	4.6.11
110.	Spherical power	3.11	4.6.12
111.	Collimation - telescope to tracker	3.8.1	4.6.13.1
112.	Collimation - checksight to tracker	3.8.2	4.6.13.2
113.	Wedge deviation, line of sight	3.13.1	4.6.13.3
114.	Wedge deviation, repeatability	3.13.2	4.6.13.4
115.	Diaphragm adjustment - elevation	3.15.1	4.6.13.5
116.	Diaphragm adjustment - azimuth	3.15.2	4.6.13.6
117.	Checksight lens	3.16	4.6.13.7
118.	Projected checksight image beam	3.18	4.6.13.8
119.	Signal frequency	3.14	4.6.13.9
120.	Irradiant intensity	3.17	4.6.13.10
121.	Operability - telescope locking lever	3.19	4.6.14
122.	Operability - checksight levers (ERROR AND ALIGN)	3.19	4.6.15
123.	Torque - ALIGN lever latching	3.19	4.6.15
124.	Operability - diaphragm adjusting	3.19	4.6.16
125.	Solenoid operation	3.19.1	4.6.17
126.	Checksight illumination	3.20.1	4.6.18
127.	Circuitry	3.20.2	4.6.19
128.	Dielectric strength	3.20.3	4.6.20
129.	Insulation resistance	3.20.4	4.6.21
130.	Workmanship	3.22	Visual - tactile

MINOR: NONE

NOTE: The tests of Table I shall be conducted at a temperature between +60° and +90°F.

4.3.3 Disposition of non-conforming product.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- One mount shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

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TABLE II

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Fabrication	3.1	Applicable drawing - visual
302.	General specification	3.2	MIL-F-13926 - visual

4.4.2 Environmental.- Three mounts shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III.

TABLE III

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
303.	Storage temperatures (-80° and +160°F)	3.3.1	4.6.1
304.	Operability - telescope locking lever	3.3.2, 3.19	4.6.2, 4.6.14
305.	Operability - checksight levers (ERROR AND ALIGN)	3.3.2, 3.19	4.6.2, 4.6.15
306.	Torque - ALIGN lever latching	3.3.2, 3.19	4.6.2, 4.6.15
307.	Operability - diaphragm adjusting screws (AZ and EL)	3.3.2, 3.19	4.6.2, 4.6.16
308.	Solenoid operation	3.3.2, 3.19.1	4.6.2, 4.6.17

NOTE: The tests for Characteristics 304 through 311 inclusive shall be conducted at -65° and +150°F.

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.1.1 Radiometer test set.- This test set will be provided upon request from Frankford Arsenal (see 6.3) to determine compliance with 3.17 and 4.6.13.10.

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4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Sealing.- The pressure testing equipment to test sealing of the mount shall conform to the internal pressure test requirements of Sheet 1 of Drawing F8565556, Pressure Tester-Two Station.

4.5.2.3.2 Orientation, collimation, wedge deviation, diaphragm adjustment, checksight lens, projected checksight, image beam.- The testing equipment utilized for orientation of the mount shall secure the telescope mounting surface of the mount in a vertical plane within 10 seconds, and the locating keyway for the telescope at the six o'clock position within 10 seconds. The testing equipment shall provide a target projector collimator that projects a calibrated target pattern into the optical system of the mount through its entrance window. The target pattern shall be projected in parallel light, unless otherwise specified. The light source for projecting the target pattern shall be capable of varied intensity, as required, when projecting the target image into the observation telescope located at the rear of the mount in line with the missile tracker cavity. The target projector shall be autocollimated to the vertical telescope mounting surface of the mount. The testing equipment shall also provide an observation telescope positioned at the rear of the mount, in line with the missile tracker cavity, for measuring the optical accuracies of the missile tracker system. The observation telescope shall be at least 10 power (10X) with at least a 2.0 inch entrance pupil, and employ an image erecting system. The telescope shall have an objective focusing capability from infinity down to 500 meters. The telescopic system shall be capable of measuring

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angles in azimuth and elevation with an accuracy of 2.0 seconds of arc. This telescope shall be autocollimated to the vertical telescope mounting surface of the mount. The testing equipment shall also provide an observation telescope positioned at the rear of the mount, in line with the telescope cavity, for measuring the optical accuracies of the telescope system. This observation telescope shall be at least 6 power (6X) with at least a 2.0 inch entrance pupil. The telescopic system shall be capable of measuring the optical accuracies of the telescope system within an accuracy of 20.0 seconds of arc. The telescope shall be adjusted for best focus of the target image. The telescope shall be autocollimated to the vertical of the telescope mounting surface of the mount. The verticle reticle line of each observation telescope and the vertical target image line shall be plumb. Autocollimation and alignment of the target projector shall be accomplished with the mount removed from the line of sight through the telescopes to target. The equipment shall be equipped with a power supply capable of energizing the chopper motor with 115 volts ac ± 6.0 volts RMS at a frequency of 400 cycles per second within ± 0.4 cycles. The testing equipment shall also be equipped with a power supply capable of energizing the checksight lamp with a voltage of 14 ± 0.5 VDC and at a current of 0.080 amp ± .002 amps and the solenoid with voltages between 18 VDC and 28 VDC. The 14 ± 0.5 VDC for the checksight lamp and the 18 VDC to 28 VDC shall be supplied from the same power source utilizing a voltage divider or other suitable method in order to have a common ground connection from one power supply.

4.5.2.3.3 Signal frequency.- The testing equipment utilized to perform this test shall be capable of recording light pulsation accuracy within ± 3 pulses to determine compliance with tolerances specified in 3.14 for light pulsations.

4.5.2.3.4 Torque requirements.- A torque wrench type tester with a minimum range from zero to twenty inch pounds shall be utilized in this test. The torque wrench type tester shall have an accuracy of plus or minus four inch ounces within its 7 to 15 inch pound measuring range. This torque wrench type tester shall be equipped with a suitable adapter to accommodate testing the latching of the ALIGN lever.

4.5.2.3.5 Test equipment set-up and calibration.- Rigidity, stability, ease of maintenance, and adaptability of set-up and calibration equipment with complete provisions are also prime requirements. All set-up and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment.

4.6 Test methods and procedures.

4.6.1 Storage temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.1. The mount shall be

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exposed to each temperature extreme specified in 3.3.1 for a minimum of 5 hours. After each temperature exposure the mount shall be stabilized in an ambient temperature between +60° and +90°Fahrenheit (F) and subjected to the visual, tactile, and audible examination of 4.6.3.1. The mount shall conform to the requirements of 3.3.1. Caution shall be exercised during this test to avoid subjecting the mount to thermal shock.

4.6.2 Extreme operating temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-E-13926 and the conditions of 3.3.2. The mount shall be exposed to each temperature extreme specified in 3.3.2 for a period of 5 hours. At the end of each 5 hour period, and while still at the temperature extreme, the mount shall be subjected to the applicable test of 4.6.14 through 4.6.17 inclusive. The mount shall conform to the requirements of 3.3.2. Caution shall be exercised during this test to avoid subjecting the mount to thermal shock.

4.6.3 Vibration.- Vibrating equipment capable of providing the amplitude and frequency specified in 3.4 shall be utilized for this test. The vibrating equipment shall also be capable of vibrating the mount along 3 mutually perpendicular axes while the mount is maintained in its using position, as depicted in the lower view on Sheet 1 of Drawing F10527950. The mount shall be horizontally mounted on the vibration table by means of an adapter. The mount shall be vibrated in accordance with Procedure IX, Part 1 of MIL-STD-810, except that the frequency, frequency cycling, and amplitude of the vibration shall be as specified in 3.4. Upon completion of the vibration cycling the mount shall be subjected to the tests specified in 4.6.3.1.

4.6.3.1 Examination.- Subsequent to vibration the mount shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine conformance with 3.4, prior to being subjected to the remaining tests specified herein. A visual examination of the mount shall be made for missing or damaged parts. Minimum visual examination shall include the windows, mirror, plate, beam-splitter, prism, wedge, tracker and missile cavities, azimuth and elevation adjusting screws, levers, mounting plate, bushings, screws, sealant, solenoid, locating pins, wiring, electrical connections and castings. A tactile examination should also be conducted to determine looseness or improper function of above items and supporting or assembled members. Optical characteristics shall be examined by observing through the entrance and exit windows to detect excessive dirt, obstruction or interference with the optical system that would be cause for rejection. An audible examination shall be made by shaking the mount while listening for loose internal optical and mechanical parts (mirror, plate, beam-splitter, prism, screws, etc.). The electrical circuit shall be inspected for damage by energizing the lamp and chopper motor, and actuating the wedge and solenoid and conducting a visual and audio examination for malfunction such as:

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failure or intermittent operation of chopper motor, flicker or intermittent illumination from lamp, binding or inoperability of wedge and solenoid and etc. Any one of the possible defects defined above shall be cause to consider a mount defective. Mounts which have passed the above examinations shall be subjected to the tests specified in 4.6.4 through 4.6.21 inclusive, to determine overall compliance with 3.4.

4.6.4 Cleanliness.- Optical surfaces of the assembled mount shall be inspected for cleanliness in accordance with the requirements of 3.5.1 and 3.5.2 utilizing the inspection techniques of MIL-O-13830 as applicable. The mount shall conform to the cleanliness requirements of 3.5.1 and 3.5.2.

4.6.5 Sealing.- The internal pressurizing of the mount to test sealing shall be performed with special testing equipment conforming to 4.5.2.3.1. Secure the poppet in valve 10516717 in a closed position. The pressure shall be applied through the valve in the mount. Pressurize the mount with dry nitrogen gas to 5.0 ± 0.10 psig in accordance with the procedure for internal pressure testing on Sheet 1 of Drawing F8565556. The dry nitrogen gas shall have a dew point at least as low as minus 25°F. The applied pressure shall be retained for a minimum of 2 hours. Within the first 60 minutes of this 2 hour time period, the ERROR lever shall be rotated through a minimum of 3 full excursions (CCW stop to CW stop to CCW stop), of its rotational range. Any evidence of leakage in the mount, at the end of the 2 hour time period, shall not exceed the tolerance specified in 3.5, as measured on the pressure gauges of the testing equipment. Upon completion of this test, release the poppet in valve 10516717.

4.6.6 Interior atmosphere.- Perform this test in sequence to the test in 4.6.5. The special testing equipment specified in 4.6.5 shall be utilized in this test. A dew-point tester, with an appropriate adapter to accommodate the Valve 10516717, shall be used to test the dew-point of the interior atmosphere of the mount. The mount shall be flushed with dry nitrogen gas. The flushing shall continue until the nitrogen gas bled from the mount has a dew point at least as low as minus 25°F, as measured with the dew point tester. The dew point tester shall be removed from the mount. The internal pressure of nitrogen gas in the mount shall be brought to a measured 3 psig ± 0.5 psig. The internal pressure of the mount shall be monitored for a period of 5 minutes ± 15 seconds. At the end of the 5 minute period there shall be no evidence of leakage in the mount as indicated on the pressure gauge of the testing equipment. The mount shall conform to the requirements of 3.21.

4.6.7 Insertion and removal (telescope and tracker).- Position the mount on holding fixture that secures its mounting surface in a vertical plane within 5 degrees. Perform this test utilizing Functional Gage, Drawing F10541523, to simulate the telescope, and Functional Gage, Drawing F10541525, to simulate the missile tracker. Insert Gage 10541523 into the telescope cavity of the mount and lock

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in place by means of the telescope locking mechanism of the mount, then insert Gage 10541525 into the tracker cavity of the mount and secure in place by means of its two mounting bolts. Remove the Gage 10541525 from the mount. During this insertion and removal of the Gage 10541525 there shall be no interference or restriction caused by the mounted Gage 10541523 or the checksight housing of the mount. Continue this test by inserting and securing the Gage 10541525 in the mount, then inserting and removing the Gage 10541523. The mount shall conform to the requirements of 3.7.2.

4.6.8 Rigidity.- Perform this test concurrent with the test in 4.6.7 utilizing the Gage 10541523. When the gage is inserted and locked in the telescope cavity of the mount, it shall be subjected to a visual and tactile examination for looseness and shake. The telescope locking mechanism of the mount shall conform to the requirements of 3.7.3.

4.6.9 Resolution.- Position the mount as specified in 4.6.7. This test shall be performed on the telescope cavity axis utilizing an auxiliary telescope of at least 24 power with a 2 inch entrance pupil, a resolving power chart (on a target or in a collimator projector) representing the angular subtense for the seconds of arc specified for resolution, and a supporting medium to receive the test equipment as well as the item to be tested. The test method and procedure shall be in compliance with the resolution test specified in MIL-O-13830, except that the focus of the auxiliary telescope shall be established at one best focus setting for simultaneous viewing of the four meridians of the resolving power chart. The resolving power chart shall contain four line sets as shown in Figure 1 of MIL-O-13830. The limit of resolution shall be within the value specified in 3.9 as determined by viewing through the auxiliary telescope.

4.6.10 Light transmission.- Light transmission shall be measured utilizing a white light source and a photocell with photometer. The measurement shall be made with the ALIGN and ERROR (wedge) levers in their maximum counterclockwise position. The light transmission as measured at the telescope cavity shall not be less than the value specified in 3.10.

4.6.11 Astigmatism.- This test shall be performed utilizing a calibrated millidioptometer and the resolving power chart specified in 4.6.9. The millidioptometer shall have a magnification of at least twenty power (20X) with a 2 inch entrance pupil, and a measuring range of at least plus or minus 6 millidiopters. Set the millidioptometer objective sleeve to zero millidiopters and adjust the millidioptometer eyepiece for best focus of the millidioptometer reticle. Position the optical axis of the millidioptometer on the mechanical axis of the telescope cavity. Best focus of each of the 4 quadrants that can be resolved by adjustment of the millidioptometer objective sleeve shall be within the tolerance (millidiopters) specified in 3.12.

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4.6.12 Spherical power.- Spherical power shall be measured with the millidioptometer oriented as in 4.6.11. Obtain best focus of the numeral in the resolving target by adjustment of the millidioptometer objective sleeve. The millidiopter value shall not exceed the tolerance specified in 3.11 as measured on the millidiopter scale.

4.6.13 Orientation.- Position the mount on the testing equipment specified in 4.5.2.3.2 and project the calibrated target image into the optical system of the mount. Adjust the intensity of the light source in the target projector collimator to present the target image to the observation telescope positioned at the tracker cavity. When a filter is used with the observation telescope positioned at the telescope cavity to compensate for the intensity of the target image it shall be calibrated for deviation error. The calibration of the filter shall be included in the optical accuracy measurements of the telescope cavity. The entrance window of the mount may be covered during the accuracy measurements of the checksight image beam for best contrast of the checksight image.

4.6.13.1 Collimation, telescope to tracker.- View through the telescope positioned at the missile tracker cavity and obtain best focus of the target image. The target image shall be super-imposed on the reticle of the telescope within the tolerance specified in 3.8.1, as measured with the telescope reticle. Record any deviation. View through the telescope positioned at the telescope cavity and obtain best focus of the target image. The target image shall be super-imposed on the reticle of the telescope within the tolerance specified in 3.8.1, as measured with the telescope reticle. Record any deviation. The two emerging lines of sight, telescope and tracker, shall be parallel within the tolerance specified in 3.8.1, as determined by the recorded values.

4.6.13.2 Collimation, checksight-to-tracker.- Perform this test with the ALIGN lever in its maximum clockwise position and the ERROR lever in the maximum counterclockwise position. Energize the chopper motor and checksight lamp. View through the telescope positioned at the missile tracker cavity and obtain best focus of the checksight beam image. The checksight beam image shall be superimposed on the telescope reticle within the tolerance specified in 3.8.2 as measured with the telescope reticle. Record the position of the checksight beam image with respect to the tracker cavity axis.

NOTE: Where the centering of the checksight beam image exceeds the tolerance of 3.8.2 by not more than 0.1 mil, the AZ or EL boresight knobs may be adjusted to center the beam image within the specified tolerance.

4.6.13.3 Wedge deviation, line of sight.- Position the ALIGN lever in its maximum clockwise position and the ERROR lever in its maximum counterclockwise position. Energize the chopper motor and checksight lamp. While viewing the checksight image beam through the telescope positioned at the missile tracker cavity, rotate the

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ERROR lever to its maximum clockwise position. The checksight beam image that is superimposed on the telescope reticle shall immediately deviate into the upper right quadrant of the telescope reticle. The amount of its deviation, from the position established in 4.6.13.2 shall not be less than, or greater than the azimuth and elevation values specified in 3.13.1, as measured with the telescope reticle.

4.6.13.4 Wedge deviation, repeatability.- Perform this test in sequence to the test in 4.6.13.3, while continuing to view the checksight image beam through the telescope at the missile tracker cavity. Rotate the ERROR lever to its maximum counterclockwise position, then rotate the lever to its maximum clockwise position a minimum of 3 times. Each time (3) the ERROR lever is in its maximum clockwise position, the checksight image beam shall deviate from the position established in 4.6.13.2 into the upper right quadrant within values specified in 3.13.2, as measured with the telescope reticle.

NOTE: Application of excessive force to the ERROR lever shall be avoided during the rotations of the lever so as to avoid damage to the lever and wedge assemblies.

4.6.13.5 Diaphragm adjustment, elevation.- Perform this test with the ALIGN lever in its maximum clockwise position and the ERROR lever in its maximum counterclockwise position. Energize the chopper motor and checksight lamp. While viewing through the telescope positioned at the missile tracker cavity, rotate the EL boresight knob to its maximum clockwise position. Record the position (mils) of the checksight image beam, as measured with the telescope reticle. Rotate the EL boresight knob to its maximum counterclockwise position. Again record the position (mils) of the checksight beam image. The recorded amount of elevation and depression adjustment of the checksight image beam, from the position established in 4.6.13.2, shall not be less than, or greater than the values specified in 3.15.1. The checksight image beam shall move in a vertical plane as specified in 3.15.1, as indicated by the vertical reticle line of the telescope.

4.6.13.6 Diaphragm adjustment, azimuth.- Perform this test with the ALIGN and ERROR levers positioned as specified in 4.6.13.5, and the chopper motor and checksight lamp energized. While viewing through the telescope positioned at the missile tracker cavity, rotate the AZ boresight knob to its maximum clockwise position. Record the position (mils) of the checksight image beam, as measured with the telescope reticle. Rotate the AZ boresight knob to its maximum counterclockwise position. Again record the position (mils) of the checksight beam image. The recorded amount of left and right azimuth adjustment of the checksight image beam, from the position established in 4.6.13.2, shall not be less than, or greater than the values specified in 3.15.2. The checksight image beam shall move in a horizontal plane as specified in 3.15.2, as indicated by the horizontal reticle line of the telescope.

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4.6.13.7 Checksight lens.- Perform this test with the checksight image beam in the position established in 4.6.13.2. Set the target projector collimator at the range specified in 3.16, and project the target image into the optical system of the mount. View through the telescope positioned at the missile tracker cavity and obtain coincidence of its horizontal and vertical reticle line with the horizontal and vertical target image lines. Rotate the ALIGN lever to its maximum clockwise position, and the ERROR lever to its maximum counterclockwise position. Energize the chopper motor and checksight lamp. Again view through the reticle boresight point of the observation telescope positioned at the tracker cavity and observe the checksight image beam. The checksight lens shall be in correct adjustment, as required in 3.16, when the checksight image beam in the center of the field is parallax free. Parallax is recognized as any apparent movement of the checksight image beam with respect to the target image, when the observer's head is moved slowly from side to side or up and down. An appropriate aperture stop for the observation telescope may be utilized for this test. Return the target projector collimator to infinity focus upon successful completion of this test.

4.6.13.8 Projected checksight image beam.- Position and orient the mount as in 4.6.13. Perform this test with the ALIGN lever in its maximum clockwise position and the ERROR lever in its maximum counterclockwise position. Energize the chopper motor and the checksight lamp. View through the observation telescope at the tracker cavity. The checksight image beam shall be positioned as specified in 4.6.13.2. While viewing through the observation telescope, the image checksight beam shall conform to the requirements of 3.18, as measured with the telescope reticle.

4.6.13.9 Signal frequency.- The signal frequency of the light pulsations through the beam-splitter shall be measured at the missile tracker mounting surface on the tracker axis, utilizing the test equipment specified in 4.6.13.10. An A.C. amplifier with a known gain and a frequency counter shall be used in conjunction with the test equipment. The power input to energize the amplifier shall be supplied from a power source, through a control device, that is capable of maintaining the necessary electrical characteristics for operation of the amplifier during this test. Position the ALIGN lever in its maximum clockwise position and the ERROR lever in its maximum counterclockwise position. Energize the chopper motor and the checksight lamp. Center the checksight beam at the recorded position established in 4.6.13.2. Energize the test equipment. The signal frequency derived from light pulsations through the beam-splitter plate, as read with the frequency counter shall conform to the requirements of 3.14.

4.6.13.10 Irradiant intensity (infrared power).- Perform this test in sequence to the test in 4.6.13.9. The effective power of the checksight beam shall be measured at the missile tracker mounting surface on the tracker axis, using the Radiometer Test Set provided by the Government (see 4.5.1.1 and 6.3). Position the ALIGN lever in

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its maximum clockwise position and the ERROR lever in its maximum counter-clockwise position. Energize the chopper motor and the checksight lamp from a controlled power source and maintain the current draw on the checksight lamp at 0.080 amperes ± 0.002 amperes during test measurements. The reading on the radiometer test set for effective power of the checksight beam shall be within the green band to be acceptable in conformance with 3.17 when the radiometer test set is energized.

4.6.14 Operability, telescope locking lever.- The telescope locking lever shall be subjected to a visual and tactile examination and shall conform to the operability requirements of 3.19.

4.6.15 Operability, checksight levers (ERROR and ALIGN).- The checksight levers (ERROR and ALIGN) shall be subjected to a tactile examination and shall conform to the operability requirements of 3.19. Standard torque measuring equipment capable of measuring the torque value specified in 3.19 shall be utilized to check the torque required to latch the ALIGN lever. An appropriate adapter to accommodate the ALIGN lever arm shall be used with the torque measuring equipment. Torque shall be measured while rotating the ALIGN lever clockwise to its latched position. The torque required to latch the ALIGN lever (full clockwise position) shall not exceed the value specified in 3.19.

4.6.16 Operability, diaphragm adjusting screws (AX and EL).- The diaphragm adjusting screws (AX and EL) shall be subjected to a tactile examination, utilizing a suitable screwdriver or fingers, to determine conformance with the operability requirements of 3.19.

4.6.17 Solenoid operation.- Connect a current measuring device in the solenoid circuitry of the mount. This device shall be capable of measuring and recording the amount of current draw (amperes) specified in 3.19.1. Turn the ALIGN lever to its maximum clockwise position. While observing the ALIGN lever and the porro prism, energize the solenoid with the minimum voltage specified. The measured current draw, the ALIGN lever, and the porro prism shall conform to the requirements of 3.19.1. Return the ALIGN lever to its maximum clockwise position. While observing the ALIGN lever and the porro prism energize the solenoid with the maximum voltage specified. The measured current draw, the ALIGN lever, and the porro prism shall conform to the requirements of 3.19.1.

4.6.18 Checksight illumination.- Connect a current measuring device in the checksight lamp circuitry of the mount. This device shall be capable of measuring the amount of current drawn (amperes) specified in 3.20.1. Turn the ALIGN lever clockwise to its full excursion of travel. Energize the chopper motor and the checksight lamp. The checksight dot image shall be clearly defined when visually examined through the exit windows of the mount. The measured current drawn shall conform to the requirements of 3.20.1..

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4.6.19 Circuitry.- A continuity check of the mount circuitry shall be made using standard measuring equipment to determine conformance with the requirements of 3.20.2.

4.6.20 Dielectric strength.- The mount shall be tested in accordance with Method 301 of MIL-STD-202, utilizing standard measuring equipment. Dielectric strength shall be in conformance with the requirements of 3.20.3. This is an in-process test that is performed with the checksight lamp, the motor, and the solenoid removed from the circuitry. The inspection records for each mount shall indicate the results of the test.

4.6.21 Insulation resistance.- Immediately after satisfactorily withstanding the dielectric strength test the mount shall be subjected to the insulation resistance test in accordance with Method 302. Test Condition B of MIL-STD-202. Measurements of all insulation between ungrounded circuits and ground, and between isolated circuits, shall be made utilizing standard measuring equipment. Insulation resistance shall be in conformance with the requirements of 3.20.4. This is an in-process test that is performed with the checksight lamp, the motor, and the solenoid removed from the circuitry. The inspection records for each mount shall indicate the results of the test.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10527950. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Telescope Mount M149 is used to support and position the telescopes M119 and M127 in the M551 Vehicle. The mount also supports the missile tracker which is part of the guidance and control equipment of the shillelagh missile. The checksight, an integral component of the mount, is used for optical alinement of the missile tracker and the telescope.

6.2 Ordering data.- Procurement documents should specify:

a. Title, data and number of this specification.
b. Order of document precedence.
c. Applicable packaging data sheet number.
d. Selection of applicable levels of preservation, and packing.
e. Availability of classified documents.
f. Submission of test data by the contractor.
g. Contractor's handling of classified test data.

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6.3 Government furnished equipment (GFE).- Request for "Radiometer Test Set - Part No. 11728050" shall be addressed as follows:

Commanding Officer
Frankford Arsenal
ATTN: Inspection Equipment Supply Branch, Q5300
Bridge & Tacony Streets
Philadelphia, Pa. 19137

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A408

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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